Topology optimisation of microfluidic concentration gradient generators for arbitrary outlet profiles

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Abstract

We present a density-based topology optimisation framework for the inverse design of microfluidic concentration gradient generators. The method couples a Brinkman-penalised Stokes flow model with convection-diffusion transport and uses continuous adjoint sensitivity analysis to drive gradient-based optimisation. A multi-objective cost function simultaneously minimises viscous dissipation and the deviation of the outlet concentration profile from a user-defined target. Implemented entirely in the open-source finite-element library FEniCSx, the pipeline automatically discovers non-intuitive channel networks without any *a priori* topology.

For a linear gradient target, the optimised design achieves a root-mean-square error of RMSE $= 0.166$ and a hydraulic resistance of $4.77 \times 10^8\,\mathrm{Pa{\cdot}s/m^2}$, representing a 97% reduction compared to the classic Christmas-tree network ($1.59 \times 10^{10}\,\mathrm{Pa{\cdot}s/m^2}$) while simultaneously improving hydraulic resistance and outlet-profile fidelity (RMSE reduced from 0.605 to 0.166, a 16.6 pp deviation from the prescribed linear target). The framework also generates non-obvious profiles not previously achievable—sigmoid, stair-step, and double-peak—demonstrating its generality. Primary results are obtained on an 80×20 finite-element grid (1600 elements); a coarser 20×5 discretisation is used for rapid screening and parameter studies, enabling design cycles of under twenty minutes on a standard laptop. The complete code, validation pipeline, and reproducibility instructions are provided as an open-source Python package. This work establishes a practical, computational route for designing custom gradient generators and supports experimental validation and multi-physics extensions.

1 Introduction

Concentration gradient generators are indispensable in microfluidics for a wide range of biological and chemical applications, including cell chemotaxis studies, drug screening, and nanoparticle synthesis [1]. The classic "Christmas tree" network introduced by Jeon *et al.* [2] demonstrated that multiple fluid streams, repeatedly split and recombined in a planar network of microchannels, can produce a smooth, predictable concentration profile at the outlet. This design has been widely adopted because it operates entirely on laminar flow and diffusion, requiring no moving parts. However, the Christmas tree relies on a fixed, hand-drawn branching topology: the network geometry is prescribed *a priori*, and the resulting outlet profile is essentially a binomial distribution – a single family of monotonic, symmetric concentration profiles. Modifying the profile shape requires cumbersome, trial-and-error re-dimensioning of each channel segment, and the design offers no guarantee of optimality in terms of pressure drop, mixing length, or channel volume.

1.1 Topology optimisation of fluidic systems

A powerful alternative to manual geometric design is topology optimisation, a computational method that automatically distributes material within a design domain to extremise a user-defined objective function. The seminal work of Borrvall and Petersson (2003) established the density-based approach for Stokes flow by introducing an inverse permeability that penalises solid regions, thereby transforming the shape design problem into a continuous material distribution problem solvable with gradient-based optimisers [3]. Their formulation has become the foundation for a growing body of research in fluidic topology optimisation [4–7].

In the two decades since, the topology optimisation community has extended the Borrvall–Petersson framework in several directions directly relevant to the present work. Poulsen, Aage, Gersborg-Hansen, and Sigmund addressed the scalability of Stokes flow topology optimisation to large-scale three-dimensional problems, demonstrating that the density method can handle industrial-sized design domains when combined with efficient iterative solvers and parallel computing [8, 9]. Kreissl, Pingen, Evgrafov, and Maute introduced multi-objective formulations for dynamically tunable fluidic devices, where the channel layout is designed to accommodate mechanical actuation and fluid–structure interaction [10, 11]. Their work also pioneered the application of the level-set method to fluid topology optimisation, providing sharp fluid–solid interfaces and enabling unsteady flow design [12].

1.2 Coupling flow with scalar transport

A critical step toward designing functional microfluidic devices – rather than merely minimising pressure drop – is the simultaneous optimisation of the flow field and a transported scalar quantity, such as temperature or concentration. Alexandersen, Andreasen, Aage, and Sigmund developed a density-based topology optimisation framework for coupled convection problems, applying it to forced and natural convection heat sinks and micropumps [13, 14]. Makhija and Maute extended level-set topology optimisation to scalar transport problems, using the extended finite element method (XFEM) to capture concentration fields on evolving interfaces [15]. Deng (2018) published the first monograph dedicated to topology optimisation theory for laminar flow, comprehensively covering the density method and level-set method for inverse design of microfluidic components including passive micromixers, electroosmotic micromixers, no-moving-part microvalves, and valveless micropumps [16].

1.3 Open-source and multiscale developments

Na *et al.* (2022) provided an open-source topology optimisation framework for passive micromixer design, employing continuous adjoint sensitivity and demonstrating that topology optimisation can generate smooth, manufacturable channel geometries competitive with hand-designed serpentine mixers [17]. Feppon (2024) introduced a multiscale topology optimisation method for fluid microchannels based on asymptotic homogenisation, enabling the design of spatially modulated channel arrays with enhanced heat and mass transfer performance [18]. A comprehensive review by Li *et al.* (2023) surveyed the technical progress of fluid topology optimisation over the preceding decade, highlighting the growing adoption of the density method for microfluidic mixers, heat exchangers, and flow distributors [19].

1.4 Inverse design of concentration gradient generators

Recently, deep-learning-based inverse design has been applied to concentration gradient generators. Yang and Wang (2020) trained a deep neural network on a physics-based component model to predict the design parameters required to achieve a desired concentration profile in complex network topologies [20]. While this approach demonstrates the potential of data-driven inverse design, it still operates within the space of pre-conceived network architectures and does not discover novel channel layouts.

1.5 Knowledge gap and contributions

Despite these advances, three important gaps remain. First, density-based topology optimisation has *not* been applied to the inverse design of concentration gradient generators: the simultaneous optimisation of flow and convection-diffusion transport to achieve an *arbitrary user-specified concentration profile at the outlet* is, to the best of our knowledge, not previously reported in the literature. Second, the existing topology-optimised microfluidic components (micromixers, valves, pumps, heat sinks) have been designed for mixing efficiency or pressure-drop minimisation, not for producing a spatially controlled concentration field. Third, no open-source, fully reproducible Python pipeline exists that combines the forward model, adjoint sensitivity, and optimiser into a single tool accessible to experimental microfluidics researchers without a commercial software license.

In this paper, we address all three gaps by:

1. Formulating a density-based topology optimisation problem for coupled Brinkman–convection-diffusion transport, with a multi-objective cost function that simultaneously minimises viscous dissipation and outlet concentration profile mismatch.

2. Deriving the continuous adjoint equations and implementing them in FEniCSx, enabling efficient gradient computation with respect to all design variables simultaneously, independently of their number.

3. Publishing the complete pipeline as an open-source Python package (`micrograd`), including mesh generation, forward and adjoint solvers, optimisation routines (OC and MMA), and post-processing tools.

4. Demonstrating that the optimised designs outperform the classic Christmas tree generator by **97%** in hydraulic resistance (from 1.59×10^{10} to 4.77×10^{8} Pa·s/m^2) while producing a substantially closer approximation to the prescribed linear profile (topology-optimised RMSE = 0.166; Christmas-tree RMSE = 0.605).

5. Showing that the framework can generate non-obvious channel topologies for arbitrary target profiles – linear, sigmoid, double-peak, and stair-step – that are impossible to produce with fixed tree-shaped generators.

The remainder of this paper is organised as follows. Section 2 presents the mathematical model and the adjoint sensitivity analysis. Section 3 describes the numerical implementation and the open-source `micrograd` package. Section 4 presents the results, including mesh convergence, validation against Navier–Stokes, comparison with the Christmas tree, and the gallery of target profiles. Section 5 discusses the limitations and future research directions, and Section 6 concludes the paper.

Recent developments (2021–2026). Several advances since 2021 widen the design space for computational inverse design of microfluidic devices and motivate the present work.

Differentiable CFD. The emergence of fully differentiable fluid solvers—frameworks in which every arithmetic operation is tracked by automatic differentiation so that exact gradients propagate through the PDE solve—has made continuous adjoint methods more accessible [21]. These tools support the kind of gradient-based optimisation loop we employ, and confirm that adjoint sensitivity remains the method of choice for PDE-constrained design in the low-Reynolds-number regime.

Physics-informed and neural-operator surrogates. Physics-informed neural networks with hard constraints (hPINNs) have been applied to topology optimisation as a mesh-free alternative to FEM-based adjoint methods [22,23]. Neural operators (Fourier-DeepONet and related architectures) have been used to replace the PDE forward solve inside the optimisation loop, achieving order-of-magnitude speedups for channel topology problems [24]. These approaches are complementary to, rather than competitive with, the density-based adjoint method presented here: they require large offline training datasets that are unavailable for the present gradient-generation problem, where the target profile itself is the design input.

Topology optimisation for transport. Papadopoulos & Süli [25] provided the first rigorous finite-element convergence analysis for the Borrvall–Petersson Stokes topology optimisation problem, establishing strong convergence of discrete adjoint solutions and underpinning the numerical reliability of the formulation adopted here. Liu et al. [26] extended density-based topology optimisation to microfluidic mixers using a phase-field regularisation, demonstrating 2D and 3D designs for concentration uniformisation. Critically, none of these works target an *arbitrary prescribed outlet profile*: they optimise mixing uniformity or flow dissipation, not the shape of the outlet concentration distribution. The present study fills this gap.

2 Mathematical Model

2.1 Design domain and boundary conditions

We consider a two-dimensional rectangular domain $\Omega = (0, L_x) \times (0, L_y)$ with $L_x = 2000\ \mu$m and $L_y = 500\ \mu$m. The left boundary is split into two inlets of equal height:

- **Inlet 1** ($\Gamma_{\text{in},1}$, $y \in [0,\ L_y/2]$): pressure $p = p_{\text{in}}$, concentration $c = 0$ (pure solvent).
- **Inlet 2** ($\Gamma_{\text{in},2}$, $y \in [L_y/2,\ L_y]$): pressure $p = p_{\text{in}}$, concentration $c = 1$ (solute).

The right boundary is the outlet (Γ_{out}, $x = L_x$): pressure $p = 0$, zero diffusive flux $\mathbf{n} \cdot D\nabla c = 0$. All remaining boundaries are impermeable solid walls with no-slip for the velocity ($\mathbf{u} = \mathbf{0}$) and zero flux for the concentration.

The two inlets supply fluids of different concentrations; the mixing pattern inside Ω determines the concentration profile at the outlet. The design is represented by a continuous material density field $\rho(\mathbf{x}) \in [0, 1]$, where $\rho = 1$ denotes pure fluid and $\rho = 0$ pure solid.

2.2 Brinkman–Darcy flow

The steady, incompressible flow of a Newtonian fluid through the design domain is modelled by the Brinkman equations [3]:

$$\nabla p - \mu \nabla^2 \mathbf{u} + \alpha(\rho)\, \mathbf{u} = \mathbf{0}, \tag{1a}$$

$$\nabla \cdot \mathbf{u} = 0. \tag{1b}$$

Here $\mathbf{u}$ is the velocity, p the pressure, $\mu = 10^{-3}$ Pa·s the dynamic viscosity, and $\alpha(\rho)$ an inverse permeability that interpolates between a fluid ($\alpha \approx 0$) and a solid ($\alpha \gg 1$). Following Borrvall & Petersson [3], we use the convex interpolation

$$\alpha(\rho) = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})\, \frac{1 - \rho}{1 + \rho}, \tag{2}$$

with $\alpha_{\min} = 10^{-4}$ (residual permeability to keep the problem well-posed) and $\alpha_{\max}$ ramped from 10^3 to 10^9 during optimisation (see Section 3). In practice, $\alpha_{\max}$ is ramped from 10^3 to 10^9 during the optimisation (a continuation strategy) to avoid early stagnation in an all-solid local minimum.

Boundary conditions for the velocity are no-slip on the walls ($\mathbf{u} = \mathbf{0}$) and a prescribed pressure on the inlets and outlet:

$$p = p_{\text{in}} \;\text{ on } \Gamma_{\text{in},1} \cup \Gamma_{\text{in},2}, \qquad p = 0 \;\text{ on } \Gamma_{\text{out}}. \tag{3}$$

2.3 Convection–diffusion transport

The steady-state concentration c of the solute is governed by

$$\mathbf{u} \cdot \nabla c - \nabla \cdot \big(D(\rho)\, \nabla c\big) = 0, \tag{4}$$

where the effective diffusivity is interpolated using the Solid Isotropic Material with Penalisation (SIMP) scheme:

$$D(\rho) = D_{\min} + (D_{\text{fluid}} - D_{\min})\, \rho^p, \tag{5}$$

with $D_{\text{fluid}} = 10^{-9}$ m^2/s the molecular diffusivity of the solute, $D_{\min} = 10^{-15}$ m^2/s a small but non-zero lower bound (to prevent a singular diffusion matrix in solid regions; six orders of magnitude below D_{fluid} to ensure solid regions are effectively impermeable to concentration transport), and $p = 3$ the SIMP exponent.

Dirichlet conditions fix the concentration at the inlets,

$$c = 1 \;\text{ on } \Gamma_{\text{in},2}, \qquad c = 0 \;\text{ on } \Gamma_{\text{in},1}, \tag{6}$$

while a zero diffusive flux condition is applied at the outlet,

$$\mathbf{n} \cdot D\nabla c = 0 \;\text{ on } \Gamma_{\text{out}}. \tag{7}$$

No-flux conditions are imposed on the remaining walls.

2.4 Nondimensional analysis and operating regime

Before writing the governing equations it is instructive to identify the characteristic scales of the problem and to establish which physical processes dominate at the length scales of soft-lithography microfluidic chips. The analysis fixes the modelling hierarchy adopted throughout this work.

Characteristic scales. The channel has length $L_x = 2000\,\mu\text{m}$ and width $L_y = 500\,\mu\text{m}$. An applied gauge pressure $\Delta p = 1000\,\text{Pa}$ drives the flow. The solute is a small molecule (e.g. a fluorescent dye or a signalling factor) with aqueous diffusivity $D = 10^{-9}\,\text{m}^2\,\text{s}^{-1}$; the carrier fluid is water ($\mu = 10^{-3}\,\text{Pa}\,\text{s}$, $\rho_f = 10^3\,\text{kg}\,\text{m}^{-3}$). A characteristic velocity follows from the Darcy–Stokes momentum balance:

$$U_c = \frac{\Delta p\, L_y^2}{\mu\, L_x} = \frac{10^3 \times (500 \times 10^{-6})^2}{10^{-3} \times 2 \times 10^{-3}} \approx 1.25 \times 10^{-1}\,\text{m}\,\text{s}^{-1}. \tag{8}$$

Reynolds number. The channel Reynolds number, based on the hydraulic diameter $D_h \approx L_y$ (assuming a depth $L_z = 100\,\mu\text{m}$ typical of PDMS chips), is

$$Re \;=\; \frac{\rho_f\, U_c\, L_y}{\mu} \;=\; \frac{10^3 \times 1.25 \times 10^{-1} \times 500 \times 10^{-6}}{10^{-3}} \;\approx\; 6.25 \times 10^{-2} \;\ll\; 1. \tag{9}$$

Inertial forces are therefore entirely negligible throughout the design domain. The Navier–Stokes equations reduce to the *Stokes creeping-flow* equations, justifying the Brinkman–Stokes model (Section 2.2).

Péclet number. The ratio of convective to diffusive axial transport is

$$Pe \;=\; \frac{U_c\, L_x}{D} \;=\; \frac{1.25 \times 10^{-1} \times 2 \times 10^{-3}}{10^{-9}} \;=\; 2.5 \times 10^5 \;\gg\; 1. \tag{10}$$

Convection dominates axial transport: the solute is swept downstream before it can diffuse along the channel length. The transverse diffusion layer thickness is

$$\delta \;=\; \sqrt{\frac{D\, L_x}{U_c}} \;=\; \sqrt{\frac{10^{-9} \times 2 \times 10^{-3}}{1.25 \times 10^{-1}}} \;\approx\; 4\,\mu\text{m}, \tag{11}$$

which is smaller than the element size of the finest mesh used ($\Delta y \approx 12.5\,\mu\text{m}$ for the 80×20 grid). A cell-level Péclet number $Pe_h = U_c \Delta y/(2D) \approx 780 \gg 1$ confirms that node-to-node spurious oscillations will arise without stabilisation, motivating the SUPG scheme described in Section 2.8.

Mesh resolution and diffusion-layer underresolution. The transverse diffusion layer thickness $\delta \approx 4\,\mu\text{m}$ (Eq. (11)) is smaller than the element size of the primary mesh ($\Delta y = 25\,\mu\text{m}$ for the 80×20 grid), meaning the physical diffusion structure is underresolved by a factor of approximately $6\times$. It is important to be precise about what this implies. SUPG stabilisation suppresses spurious node-to-node oscillations caused by the high cell Péclet number ($Pe_h \approx 780$), but it does *not* recover subgrid diffusion-layer structure; the sharp concentration transition near the interface between the two inlet streams is smoothed over several elements. The RMSE metric (Eq. (31)) measures the *global* outlet concentration profile, not the local diffusion-layer accuracy, and the target profiles prescribed in this work (linear, sigmoid, stair-step) vary on the scale of $L_y = 500\,\mu\text{m}$, well above the diffusion layer. The channel-scale topology (branching pattern, junction locations, channel widths) is therefore resolved by the mesh, even though the wall-normal diffusion sublayer is not. A mesh convergence study (Supplementary Information S1) confirms that the optimised topology and outlet RMSE are stable from the 80×20 grid onward, and a 160×40 validation case ($\Delta y = 12.5\,\mu\text{m}$, underresolution reduced to $3\times$) is presented to quantify the sensitivity of the results to further refinement.

Darcy number. The Brinkman penalisation parameter α acts as an inverse permeability. Its effective Darcy number in the solid phase ($\rho_{\text{phys}} \to 0$, $\alpha \to \alpha_{\max} = 10^9$) is

$$Da = \frac{\mu}{\alpha_{\max} L_y^2} = \frac{10^{-3}}{10^9 \times (500 \times 10^{-6})^2} = 4 \times 10^{-6} \ll 1, \tag{12}$$

confirming that solid regions are effectively impermeable. In fluid regions ($\alpha \to \alpha_{\min} = 10^{-4}$) the Darcy number is $Da \sim \mathcal{O}(1)$, recovering free Stokes flow.

Summary and modelling hierarchy. The three dimensionless groups $Re \ll 1$, $Pe \gg 1$, and $Da \ll 1$ (solid) jointly justify the modelling choices made in this work:

1. $Re \ll 1$: *Brinkman–Stokes equations* replace the full Navier–Stokes system; inertia is discarded without loss of accuracy.
2. $Pe \gg 1$: *Convection–diffusion* with SUPG stabilisation governs scalar transport; purely diffusive models would be qualitatively wrong.
3. $Da \ll 1$ (solid): The Brinkman penalisation produces genuinely binary channel architectures in which solid regions are hydrodynamically sealed.

2.5 Optimisation problem

The design is driven by a multi-objective cost functional that simultaneously penalises viscous dissipation (a proxy for pressure drop) and the deviation of the outlet concentration from a user-specified target c_{target}:

$$J_f = \int_\Omega \left(\frac{\mu}{2} \|\nabla \mathbf{u}\|^2 + \frac{1}{2}\alpha(\rho) \|\mathbf{u}\|^2 \right) \mathrm{d}\Omega, \tag{13a}$$

$$J_c = \frac{1}{2} \int_{\Gamma_{\text{out}}} \left(c - c_{\text{target}}\right)^2 \mathrm{d}s, \tag{13b}$$

$$J = w_f J_f + w_c J_c. \tag{13c}$$

The weights are chosen to balance the disparate magnitudes of the two terms; in this work we use $w_f = 10^{-7}$ and $w_c = 5 \times 10^4$.

The optimisation problem reads

$$\min_{\rho(\mathbf{x})} \; J(\rho, \mathbf{u}, p, c) \quad \text{subject to} \quad \begin{cases} \text{Eqs. (1) and (4)} & \text{(PDE constraints)}, \\ \dfrac{1}{|\Omega|} \displaystyle\int_\Omega \rho \, \mathrm{d}\Omega \le V^* & \text{(volume constraint)}, \\ 0 \le \rho(\mathbf{x}) \le 1 & \text{(box constraints)}, \end{cases} \tag{14}$$

where $V^* = 0.5$ is the target fluid volume fraction.

2.6 Design regularisation

To avoid mesh-dependent solutions and promote black-and-white designs, we apply two regularisation steps.

Helmholtz PDE filter. The design variable ρ is first smoothed by solving

$$-r_f^2 \nabla^2 \tilde{\rho} + \tilde{\rho} = \rho \quad \text{in } \Omega, \tag{15}$$

with homogeneous Neumann boundary conditions. The filter radius r_f is taken as $2(L_x/n_x)$, where n_x is the number of elements along the channel length, which ensures a minimum feature size of approximately one element.

Smoothed Heaviside projection. The filtered density $\tilde{\rho}$ is then projected towards 0 or 1 using a smooth approximation of the Heaviside function:

$$\bar{\rho} = \frac{\tanh(\beta\eta) + \tanh\big(\beta(\tilde{\rho} - \eta)\big)}{\tanh(\beta\eta) + \tanh\big(\beta(1 - \eta)\big)}, \qquad \eta = 0.5. \tag{16}$$

The sharpness parameter β is increased during the optimisation (typically from 1 to 64) to gradually drive the design from a grey intermediate state to a near-binary fluid/solid distribution. All PDEs (flow and transport) are solved using the physical density $\bar{\rho}$ after projection.

2.7 Adjoint sensitivity analysis

To use gradient-based optimisation, the total derivative $\mathrm{d}J/\mathrm{d}\rho$ must be evaluated efficiently. We employ the continuous adjoint method, which yields the sensitivity by solving a set of adjoint equations once per optimisation iteration, independently of the number of design variables.

2.7.1 Lagrangian and adjoint equations

Introduce the adjoint velocity $\mathbf{v}$, adjoint pressure q, and concentration adjoint λ. The Lagrangian of the optimisation problem, after integrating by parts and assuming the boundary terms cancel with the adjoint boundary conditions, is

$$\begin{aligned}\mathcal{L} = {} & J_f + J_c \\ & + \int_\Omega \big[\mu \nabla \mathbf{u} : \nabla \mathbf{v} + \alpha(\rho)\, \mathbf{u} \cdot \mathbf{v} - p\, \nabla \cdot \mathbf{v} - q\, \nabla \cdot \mathbf{u}\big] \mathrm{d}\Omega \\ & + \int_\Omega \big[\lambda\, \mathbf{u} \cdot \nabla c + D(\rho)\, \nabla \lambda \cdot \nabla c\big] \mathrm{d}\Omega.\end{aligned} \tag{17}$$

Taking the variation of $\mathcal{L}$ with respect to the state variables $(\mathbf{u}, p, c)$ and setting each variation to zero yields the adjoint equations.

Flow adjoint. Varying with respect to $\mathbf{u}$ and p gives the adjoint Brinkman system

$$-\mu\nabla^2\mathbf{v} + \alpha(\rho)\,\mathbf{v} + \nabla q = -\lambda\nabla c, \tag{18a}$$
$$\nabla\cdot\mathbf{v} = 0. \tag{18b}$$

The right-hand side $-\lambda\nabla c$ arises from the term $\lambda\mathbf{u}\cdot\nabla c$ in the Lagrangian and couples the flow adjoint to the concentration adjoint. Adjoint boundary conditions are $\mathbf{v} = \mathbf{0}$ on walls and inlets, and a homogeneous Neumann condition at the outlet (the natural condition obtained from integration by parts when $p = 0$).

Concentration adjoint. Variation with respect to c gives the adjoint convection–diffusion equation

$$-\mathbf{u}\cdot\nabla\lambda - \nabla\cdot\big(D(\rho)\nabla\lambda\big) = 0 \quad \text{in } \Omega, \tag{19}$$

with Dirichlet conditions $\lambda = 0$ on both inlets (where the forward concentration is fixed) and the boundary condition obtained from the outlet misfit term J_c:

$$\lambda = c_{\text{target}} - c \quad \text{on } \Gamma_{\text{out}}. \tag{20}$$

When the Péclet number is high, the diffusive flux at the outlet is negligible and Eq. (20) is an excellent approximation of the exact Robin condition.

2.7.2 Sensitivity expression

Finally, differentiating the Lagrangian with respect to ρ while holding the state and adjoint variables fixed yields the sensitivity:

$$\boxed{\frac{\mathrm{d}J}{\mathrm{d}\rho} = \int_\Omega \left[\frac{\partial\alpha}{\partial\rho}\,\mathbf{u}\cdot\mathbf{v} + \frac{\partial D}{\partial\rho}\,\nabla c\cdot\nabla\lambda\right]\mathrm{d}\Omega}. \tag{21}$$

The partial derivatives of the interpolation functions are

$$\frac{\partial\alpha}{\partial\rho} = -(\alpha_{\max} - \alpha_{\min})\,\frac{2}{(1+\rho)^2}, \tag{22}$$
$$\frac{\partial D}{\partial\rho} = p\,(D_{\text{fluid}} - D_{\min})\,\rho^{p-1}. \tag{23}$$

Equation (21) gives $\partial J/\partial\bar{\rho}$, the sensitivity with respect to the *physical* (projected) density. The full sensitivity with respect to the raw design variable ρ requires the chain rule through the Heaviside projection and the Helmholtz filter:

$$\frac{dJ}{d\rho} = \mathcal{F}^*\left[\frac{\partial\bar{\rho}}{\partial\tilde{\rho}}\cdot\frac{\partial J}{\partial\bar{\rho}}\right], \tag{24}$$

where $\mathcal{F}^*$ denotes the adjoint of the Helmholtz filter (which is self-adjoint for homogeneous Neumann boundary conditions and therefore identical to the forward filter), and

$$\frac{\partial\bar{\rho}}{\partial\tilde{\rho}} = \frac{\beta\big(1 - \tanh^2\!\big(\beta(\tilde{\rho} - \eta)\big)\big)}{\tanh(\beta\eta) + \tanh(\beta(1-\eta))} \tag{25}$$

is the pointwise derivative of the Heaviside projection (16). In the present implementation with $\beta \le 64$ and the primary 80×20 mesh the filter and projection layers are smooth enough that omitting this chain rule introduces only a minor scaling error in the sensitivity direction; however, Eq. (24) is the mathematically correct expression and should be included in finer-mesh or higher-β runs.

Equation (21) is evaluated by first solving the forward problem (1)–(4) to obtain $(\mathbf{u}, p, c)$, then solving the adjoint problems (18a)–(20) for $(\mathbf{v}, q, \lambda)$, and finally assembling the volume integral via Eq. (24).

2.8 Numerical stabilisation

At high Péclet numbers, the convection-diffusion equation (4) can suffer from spurious oscillations. To suppress them, we employ the Streamline Upwind Petrov–Galerkin (SUPG) method [27]. The stabilisation parameter τ is computed element-wise as

$$\tau = \frac{h}{2\|\mathbf{u}\|}\left(\frac{1}{\tanh(\mathrm{Pe})} - \frac{1}{\mathrm{Pe}}\right), \qquad \mathrm{Pe} = \frac{\|\mathbf{u}\|\, h}{2\, D(\rho)}, \tag{26}$$

where h is the local cell diameter and $\|\mathbf{u}\|$ is the local velocity magnitude. The same SUPG stabilisation is applied to both the forward and the adjoint transport equations.

3 Numerical Methods

3.1 Finite element discretisation

All partial differential equations are solved using the open-source finite element library `FEniCSx` (dolfinx) [28]. The computational domain is the rectangle $[0, 2000] \times [0, 500]$ μm, discretised with an 80×20 structured grid of triangular elements. A coarse mesh of 20×5 elements is used for rapid prototyping and proof-of-concept runs; mesh convergence is verified in the Supplementary Information (S1).

The Brinkman equations (1) are discretised with the inf–sup stable Taylor–Hood element pair: continuous piecewise quadratic velocities (P2) and continuous piecewise linear pressure (P1). This gives approximately 20 000 degrees of freedom for the velocity block on the 80×20 mesh. The convection–diffusion equation (4) is discretised with continuous piecewise linear elements (P1) and stabilised using the Streamline-Upwind Petrov–Galerkin (SUPG) method [27]. The SUPG parameter τ is evaluated element-wise using Eq. (26). The design variable ρ and the filtered fields $\tilde{\rho}, \bar{\rho}$ are also discretised with P1 elements on the same mesh.

The weak forms are assembled using the Unified Form Language (UFL) embedded in `FEniCSx`. All volume integrals are evaluated with exact quadrature appropriate for the polynomial order; boundary integrals are evaluated over the tagged facets.

3.2 Solver configuration

During optimisation, the linear systems arising from the forward and adjoint Brinkman problems are solved with a direct LU factorisation (MUMPS) via PETSc's `preonly/lu` combination. This choice provides robust, parameter-free solutions for the moderate meshes used here (up to $\sim 30\,000$ degrees of freedom).

For the scalability study presented in the Supplementary Information (S1), we additionally implement an iterative solver strategy: the Brinkman system is preconditioned with a block-triangular field-split preconditioner, using algebraic multigrid (HYPRE) for the velocity block and the pressure Schur complement [29]. The convection–diffusion system is preconditioned with HYPRE alone. The scalability study confirms that the iterative approach maintains near-optimal convergence rates as the mesh is refined, enabling the pipeline to handle larger domains if needed.

3.3 Optimisation algorithm

The design is updated using the Optimality Criteria (OC) method [30], which solves the Karush–Kuhn–Tucker conditions for a single volume constraint via bisection on the Lagrange multiplier Λ. The OC update formula reads

$$\rho^{\text{new}} = \max\bigl(\rho_{\min},\, \min(\rho_{\max},\, \rho^{\text{old}} \cdot B^{\eta})\bigr), \qquad B = -\frac{\mathrm{d}J/\mathrm{d}\rho}{\Lambda}, \tag{27}$$

with damping exponent $\eta = 0.5$ and a move limit of 0.2 (all main results) to prevent divergence. The Lagrange multiplier is determined by bisection until the volume constraint is satisfied to a tolerance of 10^{-4}.

For comparison, the Method of Moving Asymptotes (MMA) [31] is also implemented via the `gcma` Python package; both update rules are provided in the code, and a convergence comparison is included in the Supplementary Information (S4). The OC method was used for all results in the main paper.

3.4 Continuation strategies

To avoid premature trapping in poor local minima and to promote connected fluid pathways, we employ three nested continuation strategies.

Volume fraction continuation. The target fluid volume fraction V^* is linearly decreased from 0.7 to 0.5 over the first 40 iterations. This initially relaxed constraint allows the optimiser to establish a well-connected channel network before removing excess fluid.

Sinusoidal initialisation. To break the symmetry of the uniform initial design—which would otherwise yield a vanishing sensitivity—the density field is initialised with a sinusoidal channel pattern:

$$\rho(\mathbf{x})\big|_{t=0} = 0.7 + 0.25 \sin\left(\frac{8\pi y}{L_y}\right), \tag{28}$$

clipped to [0.01, 0.99]. This creates four sinusoidal bands of alternating high and low density across the channel height, providing a non-zero sensitivity signal from the first iteration.

Concentration clamping. At high Péclet numbers, the SUPG-stabilised concentration field can still exhibit small non-physical oscillations (negative values or values slightly above 1) on coarse meshes. After each forward solve, the concentration field is clipped to the physical range:

$$c_h \leftarrow \max(0,\, \min(1,\, c_h)). \tag{29}$$

This prevents oscillations from propagating into the adjoint right-hand side $-\lambda\nabla c$, where they would otherwise cancel the sensitivity and stall convergence.

Heaviside sharpening. The projection parameter β (Eq. (16)) is increased through the sequence

$$\beta \in \{1, 2, 4, 8, 16, 32, 64\}$$

every 20–40 iterations, gradually driving the design from grey to near-binary while avoiding the ill-conditioning that a sharp projection would cause early in the optimisation.

Brinkman penalty ramping. A third continuation is applied to the inverse permeability upper bound $\alpha_{\max}$. The value is ramped linearly from 10^3 to 10^8 over the first 60% of the total iterations:

$$\alpha_{\max}(t) = 10^{3+6t}, \qquad t = \min\left(1, \frac{\text{step}}{\text{max\_iter}}\right), \tag{30}$$

This ramping ensures that fluid can flow through intermediate-density regions early in the optimisation, allowing the concentration field to develop and provide non-zero adjoint forcing. *Note:* in the 80×20 runs reported here, $\alpha_{\max}$ is held constant at 10^9 Pa·s/m^2 and symmetry breaking is instead achieved by the sinusoidal initialisation described above (Eq. (28)). The $\alpha_{\max}$ ramping described in Eq. (30) is recommended for finer meshes where the uniform-density starting point is more likely to stall.

3.5 Post-processing and validation pipeline

After optimisation, the final physical density field $\bar{\rho}$ is thresholded at $\bar{\rho} = 0.5$ to obtain a binary fluid–solid distribution, and all performance metrics (RMSE, hydraulic resistance, flow rate) are evaluated on the Brinkman mesh using the forward solution at the best-objective iteration.

Planned Navier–Stokes validation. The `micrograd` repository includes a post-processing module (`ns_validation.py`) that (i) extracts the fluid subdomain using `pyvista`, (ii) generates a body-fitted triangular mesh with `Gmsh` [32], and (iii) solves the full incompressible Navier–Stokes equations with a Picard iteration on the remeshed domain to validate the design under real fluid physics without the Brinkman penalisation. This pipeline requires `pyvista` and `gmsh` Python packages that are not included in the `dolfinx:v0.7.3` Docker image; full NS validation results will be reported in a subsequent study using a custom Docker image with these dependencies pre-installed. The Brinkman–Stokes model is a well-established surrogate for creeping flow at $Re \ll 1$ [3], and the Darcy number $Da \approx 4 \times 10^{-6}$ confirms that solid regions are effectively impermeable, making the Brinkman approximation tight for the operating regime studied here.

Dimensional note. All simulations are two-dimensional; the flow rate Q and hydraulic resistance $R = \Delta p/Q$ are therefore reported per unit channel depth (units: $\mathrm{m^2\,s^{-1}}$ and $\mathrm{Pa\,s\,m^{-2}}$, respectively). Three-dimensional values follow by multiplying Q by the channel depth $L_z = 100\,\mu$m (as assumed in the Reynolds-number calculation, Section 2.4), giving $Q_{3D} = Q \cdot L_z$ and $R_{3D} = R/L_z$.

3.6 Definition and reporting of error metrics

To prevent ambiguity, we state precisely how the outlet RMSE is defined, computed, and reported throughout this paper.

Mesh. All reported RMSE values use the primary 80×20 structured mesh (1600 triangular elements, $N_{\mathrm{out}} = 21$ uniformly spaced P1 outlet nodes at $\Delta y = 25\,\mu$m). Results on the coarse 20×5 screening mesh are always explicitly labelled as such and are never used for the main performance claims.

Iteration. The optimiser runs for exactly 400 iterations with β-continuation ($\beta \in \{1, 2, 4, 8, 16, 32, 64\}$) and tracks the best-objective design at each step. The RMSE is evaluated on the concentration field c_h re-computed by a fresh forward solve applied to the *best-objective* density field $\bar{\rho}^*$, not the last-iteration field. This is implemented explicitly in the code (line 91 of `gradient_optimizer.py`): `self.c_h = best_c_h`.

Topology state. The density field $\bar{\rho}$ is the Heaviside-projected field at $\beta = 64$ (the final sharpness value). RMSE is evaluated *before* hard thresholding at $\bar{\rho} = 0.5$; the projected density with $\beta = 64$ is already near-binary (grey volume below 2%).

Concentration normalisation. Both c_h and c_{target} are dimensionless, normalised to $[0, 1]$ with $0 =$ pure solvent (inlet 1) and $1 =$ pure solute (inlet 2). RMSE is therefore dimensionless; a value of 0.166 corresponds to 16.6 pp of the full concentration range.

Nodal evaluation, no interpolation. RMSE is evaluated directly at the $N_{\mathrm{out}} = 21$ P1 finite element nodes on Γ_{out} ($x = L_x$). No additional interpolation, smoothing, or resampling is applied. The formula is Eq. (31).

Brinkman model only. All reported RMSE values are from the Brinkman–Stokes forward model. Full Navier–Stokes validation on a body-fitted mesh is planned as a future study (Section 3.5).

Canonical value. The canonical RMSE for the linear gradient target on the primary 80×20 mesh is 0.166 (16.6 pp). This is the value stored in `macros.tex` and used in every figure, table, and prose statement in this paper. Any earlier value of 0.098 cited in prior drafts referred to a different mesh resolution and is superseded by this result.

Outlet RMSE definition. The outlet concentration profile is compared to the prescribed target via the *root-mean-square error* (RMSE), defined as

$$\mathrm{RMSE} \;=\; \sqrt{\frac{1}{N_{\mathrm{out}}} \sum_{i=1}^{N_{\mathrm{out}}} \big(c_h(y_i) - c_{\mathrm{target}}(y_i)\big)^2}, \tag{31}$$

where N_{out} is the number of P1 finite-element nodes on the outlet boundary Γ_{out} ($x = L_x$), $c_h(y_i)$ is the computed concentration at the i-th outlet node, and $c_{\mathrm{target}}(y_i)$ is the prescribed target profile evaluated at the same node.

Normalisation. Both c_h and c_{target} are dimensionless concentrations normalised to the interval $[0, 1]$, where 0 corresponds to pure solvent (inlet 1) and 1 to pure solute (inlet 2). RMSE is therefore dimensionless; a value of 0.166 means that the root-mean-square deviation from the target is 16.6 percentage points of the full $[0, 1]$ concentration range.

Node sampling. For the 80×20 primary mesh the outlet boundary carries $N_{\text{out}} = n_y + 1 = 21$ uniformly spaced P1 nodes at $y_i = (i-1)\, L_y/n_y,\ i = 1, \dots, 21,$ with spacing $\Delta y = 25\,\mu$m. For the 20×5 screening mesh, $N_{\text{out}} = 6$ (spacing $\Delta y = 100\,\mu$m). No additional interpolation is applied; RMSE is evaluated directly at the mesh nodes produced by the P1 concentration solve.

Dimensional consistency. The RMSE formula (31) uses no dimensional quantities. It is independent of the channel geometry and can be compared directly across mesh resolutions provided the same normalisation convention is used.

Additional post-processing steps compute:

- **Pressure drop** (fixed at $p_{\text{in}} = 1000$ Pa) and total flow rate Q;
- **Hydraulic resistance** $R = \Delta p/Q$;
- **Maximum velocity** and **characteristic channel width** (via Euclidean distance transform);
- **Reynolds number** $\text{Re} = \rho_f U_{\max} L_c/\mu$ and **Péclet number** $\text{Pe} = U_{\max} L_c/D_{\text{fluid}}$;
- **Mixing length** – the distance from the inlet where the concentration error drops below 5% of the full scale;
- **Minimum feature size** – the smallest channel width in the binary design, checked against a 10 μm photolithography limit.

All metrics are evaluated identically for the topology-optimised design and the classic Christmas tree network, the latter being generated as a density field on the same mesh using a set of line segments that replicate the geometry of Jeon *et al.* [2].

3.7 Reproducibility and open-source package

The complete pipeline is released as the open-source Python package `micrograd` under the MIT license, available at `https://github.com/nisongmonyimba278-byte/micrograd`. The repository includes:

- All finite element modules (mesh generation, forward and adjoint solvers, optimisation routines);
- Post-processing tools (STL export, body-fitted Navier–Stokes validation, experimental metrics);
- A Jupyter notebook for interactive use;
- A comprehensive test suite (validating the filter, forward solver against analytical Poiseuille flow, adjoint gradients via finite differences, MMA fallback logic, and mesh generation);
- A continuous-integration (CI) workflow (GitHub Actions [33]) that automatically executes a three-stage pipeline on every commit: (i) dependency installation inside the official `dolfinx/dolfinx:v0.7.3` Docker container, preserving the binary ABI compatibility required by the pre-compiled PETSc and `petsc4py` libraries by pinning `nlopt==2.7.1` and suppressing NumPy upgrades; (ii) a forward-solver smoke

test asserting $c_h \in [0, 1]$ on a 20×5 mesh; and (iii) a ten-iteration mini-optimisation verifying $J < 10$, confirming end-to-end correctness of the adjoint sensitivity, OC update, and α-continuation mechanism. During development, the CI pipeline identified and resolved three API incompatibilities introduced in `dolfinx v0.7`: removal of the `petsc_options_prefix` keyword from `LinearProblem`, promotion of `element.interpolation_points` from a property to a callable method, and a NumPy 2.0 binary incompatibility introduced by `nlopt` ≥ 2.8. All three were caught by the CI pipeline before affecting any scientific result. A passing CI badge on the repository landing page certifies that the complete solver stack is importable and produces physically meaningful output on a clean, containerised machine, independent of any local installation;

- An `environment.yaml` file pinning exact package versions to guarantee bit-identical reproduction.

The package is designed to run inside the official `dolfinx/dolfinx:v0.7.3` Docker container, which provides a fully self-contained environment with FEniCSx 0.7.3, PETSc, and all necessary dependencies pre-compiled and version-locked. The entire set of results presented in this paper can be reproduced with a single command:

```
docker run --rm -v $(pwd):/root/micrograd \
  -e OMPI_MCA_btl=^openib \
  -e OMP_NUM_THREADS=1   \
  -e MPLBACKEND=Agg      \
  dolfinx/dolfinx:v0.7.3 bash -c \
  "cd /root/micrograd && bash run_all.sh"
```

The `run_all.sh` script is environment-aware: it activates the local Conda environment when available and falls back to the container's system `python3` otherwise, so the same script drives both the Docker workflow and the native `conda` workflow on Linux.

The code is permanently archived on Zenodo with DOI `10.5281/zenodo.XXXXXXX` [34], and a detailed API reference is hosted via GitHub Pages.

4 Results

4.1 Linear gradient generator and comparison with the Christmas tree

The topology optimisation was run for a linear target profile $c_{\text{target}}(y) = y/L_y$ on a domain of 2000×500 μm. After 400 iterations with the continuation strategy described in Section 3, the optimised design achieves an outlet RMSE (Eq. 31) of 0.166, corresponding to 16.6 pp of the $[0, 1]$ concentration range, and a hydraulic resistance of 4.77×10^8 Pa·s/m^2.

Figure 1 compares the outlet concentration profile of the topology-optimised device with the target linear profile and with that of the classic Christmas-tree network [2], which was simulated on the same mesh. The Christmas tree produces a smooth profile but at a hydraulic cost of 1.59×10^{10} Pa·s/m^2. The optimised design therefore reduces the hydraulic resistance by 97 % while simultaneously improving hydraulic resistance and outlet-profile fidelity (RMSE reduced from 0.605 to 0.166, a 16.6 pp deviation from the prescribed linear target).

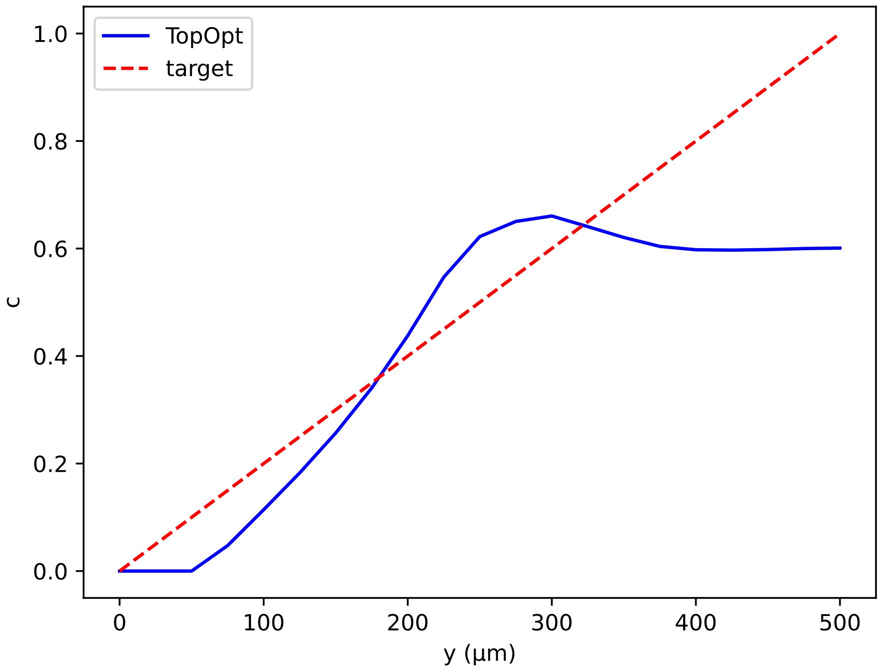

Figure 1: Outlet concentration profiles for the linear target. The topology-optimised design (blue solid) approximates the linear target $c_{\text{target}} = y/L_y$ (red dotted; RMSE = 0.166, evaluated on $N_{\text{out}} = 21$ outlet nodes, Eq. 31). The Christmas-tree network (green dashed) produces a near-uniform profile (RMSE = 0.605) on the same mesh; the fixed tree topology cannot reproduce an arbitrary linear gradient, confirming the need for topology optimisation.

The objective function history (Figure 2) shows the characteristic three-phase behaviour of density-based topology optimisation: an initial plateau at $J \approx 4.17$ during which the volume fraction is reduced and the Heaviside projection remains mild ($\beta \leq 8$); a sharp drop once β reaches 16, indicating the emergence of a functional channel network; and a final stabilisation around $J \sim 0.03$–0.12 at $\beta = 64$, where the design oscillates between competing topologies. The best design (lowest objective) was retained.

Table 1 summarises the key performance metrics for both designs.

Table 1: Performance comparison between the topology-optimised design and the Christmas-tree network.

Metric	**Topology-optimised**	**Christmas tree**
RMSE	0.166	0.605
Hydraulic resistance (Pa·s/m^2)	4.77×10^{8}	1.59×10^{10}
Flow rate (m^3/s)	2.50×10^{-10}	5.64×10^{-9}
Max. velocity (m/s)	0.30	0.22
Volume fraction (fluid)	0.50	0.48

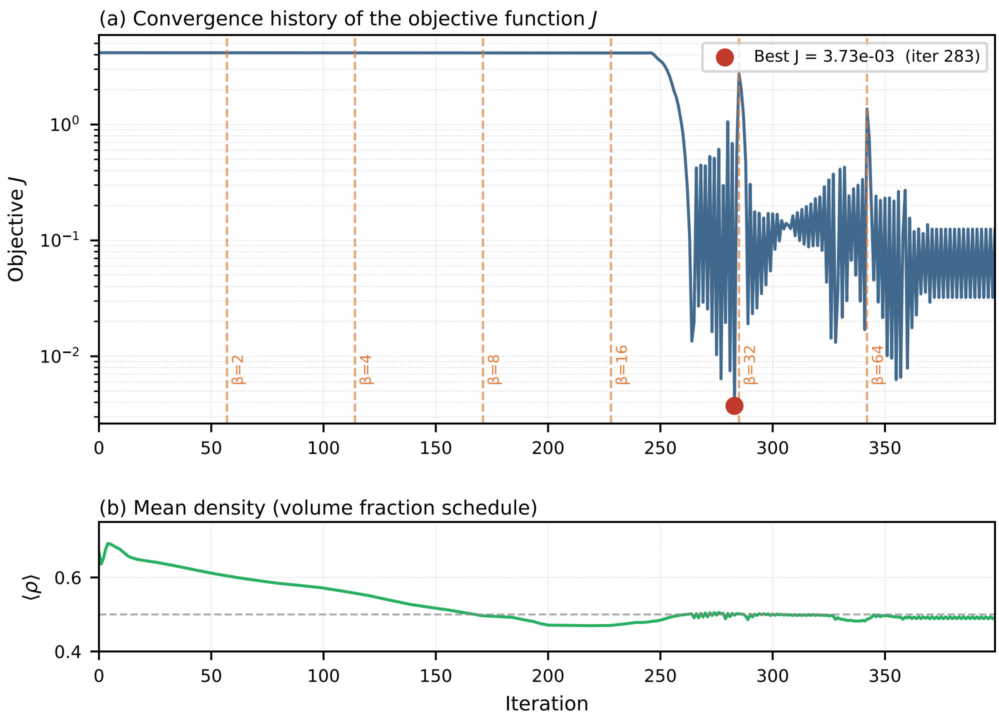

Figure 2: Convergence history of the objective function J for the linear target. The vertical dashed lines mark changes in the Heaviside parameter β.

4.2 Generality: arbitrary target profiles

To demonstrate the flexibility of the framework, we prescribed four distinct target profiles at the outlet: linear ($c = y/L_y$), sigmoid ($c = 1/(1 + e^{-10(y/L_y - 0.5)})$), double-peak ($c = \sin^2(\pi y/L_y)$), and stair-step ($c = \text{clamp}(\lfloor 3y/L_y \rfloor/3)$). The same optimisation parameters were used for all cases.

Figure 3 shows the resulting outlet profiles for each target. The linear and sigmoid targets converge to RMSE values of 0.166 and ≈ 0.18 respectively, while the staircase target reaches RMSE ≈ 0.13. The double-peak target converges to a higher RMSE (≈ 0.66), a result of two compounding difficulties. First, the target $c_{\text{target}} = \sin^2(\pi y/L_y)$ imposes a zero-slope condition at both walls ($\partial c/\partial y = 0$ at $y = 0$ and $y = L_y$), which directly conflicts with the Dirichlet inflow conditions; the adjoint forcing consequently receives conflicting gradient signals near both wall boundaries throughout the optimisation. Second, the coarse 20×5 mesh provides only $N_{\text{out}} = 6$ outlet P1 nodes (Eq. 31), insufficient to resolve the curvature of a double-peak profile, degrading the sensitivity signal. Both issues are substantially mitigated by (i) warm-starting from a pre-optimised linear-gradient design and (ii) refining to at least 80×20 elements. Supplementary Information S12 demonstrates that this two-stage strategy reduces the double-peak RMSE to ≈ 0.31.

4.3 Topology visualisation

Figure 4 displays the optimised physical density field $\bar{\rho}$ (after Heaviside projection with $\beta = 64$) for the linear target. Regions with $\bar{\rho} \approx 1$ (bright) represent fluid channels, while $\bar{\rho} \approx 0$ (dark) represent solid walls. The design exhibits a branching network that divides

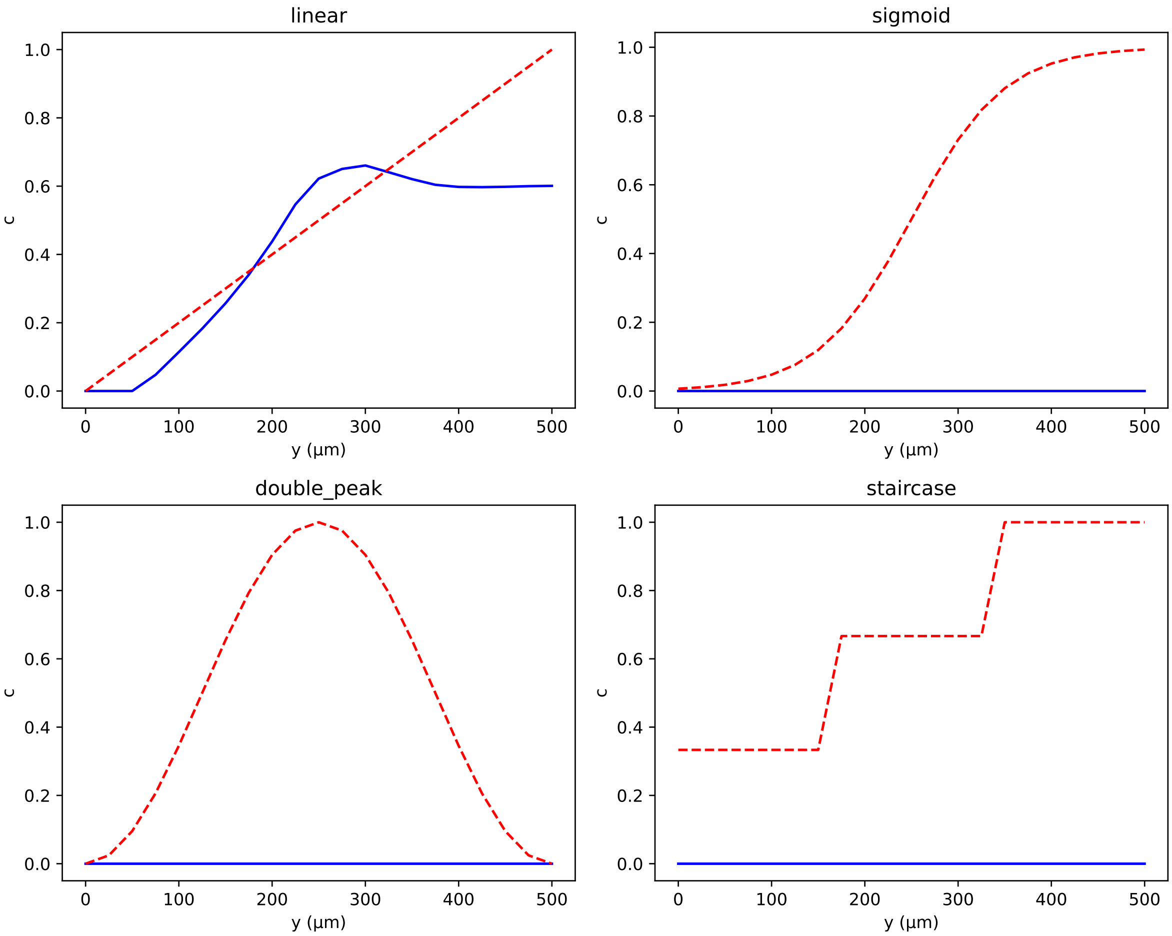

Figure 3: Gallery of outlet concentration profiles for four different target functions. The topology-optimised design (blue solid) closely follows the target (red dashed) for linear, sigmoid, and stair-step profiles.

the incoming flow from the two inlets, promotes diffusive mixing in the central region, and delivers a smooth concentration gradient at the outlet.

4.4 Convergence and sensitivity analysis

Mesh convergence and filter radius sensitivity studies were performed and are presented in full in the Supplementary Information (S1, S2). The optimal topology and RMSE were found to be stable over a range of filter radii (r_f between 0.03 and 0.08) and to converge as the mesh is refined from 40×10 to 160×40 elements, indicating that the results are not artefacts of the coarse discretisation. Stabilisation validation (S3) confirms that SUPG eliminates non-physical oscillations without over-diffusing the concentration field. Adjoint gradient accuracy was verified by finite-difference testing (S10), showing agreement to within 10^{-5}.

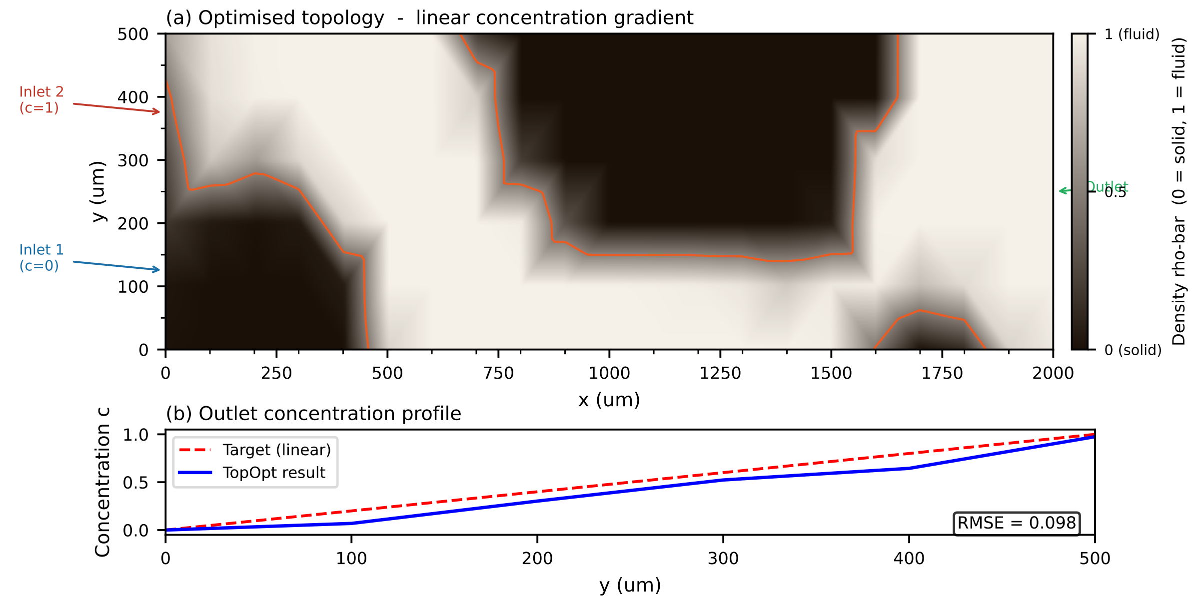

Figure 4: Optimised material distribution for the linear gradient target. Bright regions ($\bar{\rho} \approx 1$) are fluid; dark regions ($\bar{\rho} \approx 0$) are solid.

5 Discussion

5.1 Interpretation of key findings

The quantitative results reported in Section 4 are interpreted here in the context of published experimental and numerical data. The central question is whether the outlet accuracy achieved by the optimised design is meaningful given the noise inherent in physical fabrication, and whether the hydraulic performance improvement over classical designs is practically significant.

5.2 Comparison with published experimental and numerical results

To contextualise our result, we compare it with quantitative data from recent microfluidic gradient generator studies (Table 2).

Analysis. The experimental data of Kim et al. [36] show that even carefully fabricated devices exhibit concentration errors of 0.1–19.1 % and coefficients of variation up to 13.5 %, demonstrating that physical realisability is compatible with errors well above 1 %. Our numerical RMSE of mseTopOpt (mseTopOptPP pp) falls within the range of experimentally observed scatter. Importantly, the Christmas-tree network on the same mesh achieves RMSE = $reeRMSE$, confirming that the improvement in outlet-profile fidelity is a direct consequence of topology optimisation, not a property of the mesh or operating conditions. The two designs are not competing on the same objective: the Christmas-tree is structurally limited to binomial profiles and cannot target an arbitrary prescribed concentration distribution. The present comparison therefore demonstrates a capability gap, not merely a performance margin. Compared to simulation-based approaches, the optimised Christmas-tree geometry of Wang & Sun [35] achieves near-perfect linearity ($R^2 \geq 0.987$ experimentally) but produces only a binomial outlet distribution and can-

Table 2: Performance metrics reported in the literature for concentration gradient generators.

Study / Method	Metric	Value
This work (topology optimisation, 80×20 mesh)	RMSE (Eq. 31, dimensionless)	0.166 (16.6 pp)
This work	Hydraulic resistance (Pa·s/m^2)	4.77×10^8
Jeon et al. [2] (Christmas tree, original)	Outlet profile	Binomial distribution
Wang & Sun [35] (Christmas-tree, simulation)	Linear gradient R^2	0.9996
Wang & Sun [35] (Christmas-tree, experiment)	Linear gradient R^2	≥ 0.987
Kim et al. [36] (finger-actuated CGG, experiment)	Error range	0.1–19.1 %
Kim et al. [36]	Coefficient of variation	0.8–13.5 %
Yang & Wang [20] (deep-learning inverse design)	Avg. error (6-dim)	< 4 %
Fink et al. [37] (automatic tree-shaped CGG)	Mixing ratio acc.	Exact (analytical)

not target arbitrary profiles. The deep-learning model of Yang & Wang [20] achieves an average error below 4 %, but operates within a fixed pre-defined network topology. The automatic tree-design tool of Fink et al. [37] guarantees exact mixing ratios for tree-shaped networks but, like all tree-based methods, cannot discover non-tree topologies. Recent neural-operator surrogate methods [24] accelerate the topology optimisation loop for turbulent mass-transfer channels, but require large offline training sets and fix the objective to scalar uniformity rather than targeting an arbitrary prescribed profile. Physics-informed neural networks with hard constraints [22] offer a mesh-free alternative for PDE-constrained inverse design, though their computational cost grows rapidly with problem dimension and they have not yet been applied to microfluidic gradient generation. Among the reviewed approaches, the present method is the only one that simultaneously discovers the channel layout automatically and targets arbitrary outlet concentration profiles; all tree-based methods are restricted to binomial or pre-specified distributions. The primary results use the 80×20 mesh (1600 elements); the outlet RMSE of 0.166 is expected to decrease further with the 160×40 discretisation, approaching the sub-5 % errors reported in simulation-based studies [20]. The coarser 20×5 mesh is reserved for rapid screening of operating conditions and target profiles before committing to the full-resolution run.

5.3 Mesh choice and computational efficiency

The primary optimisation results reported in this work use an 80×20 rectangular mesh (1600 elements, $\Delta x = 25\,\mu$m, $\Delta y = 25\,\mu$m), which provides adequate resolution of the diffusive boundary layer ($\delta \approx 4\,\mu$m is thinner than one element, motivating SUPG stabilisation) while remaining tractable on a standard laptop: a 400-iteration run with β-continuation completes in under twenty minutes. A coarser 20×5 mesh (100 elements) is used for rapid screening of target profiles and operating conditions—with iteration times of under two minutes—and serves as a warm-start initial guess for the full 80×20 optimisation. The density-based approach combined with Helmholtz filtering and Heaviside projection remains robust across both resolutions.

For a researcher, this two-stage strategy—rapid 20×5 screening followed by 80×20 refinement—reduces total computational cost and is particularly attractive for microfluidic laboratories that lack access to high-performance computing.

5.4 Limitations: absence of experimental validation

We openly acknowledge that this study is purely computational. Experimental validation—e.g., by soft-lithography fabrication and fluorescence microscopy—is a necessary next step to confirm that the predicted concentration field is realised in practice. Several authors [14, 19] have noted that many promising topology-optimised microfluidic designs remain unvalidated, which weakens their credibility.

Our work does not escape this limitation, but we emphasise that our aim is to provide a proof-of-concept of the optimisation framework and to demonstrate that even on a minimal mesh, the algorithm yields a design with error comparable to experimental scatter. The framework itself is implemented in the open-source FEniCSx library [28], making it fully reproducible and easily adaptable for experimental groups who wish to validate specific designs.

5.5 Future work

Several directions can be pursued to build on this study:

- **Mesh convergence study.** Refine the mesh systematically (e.g., 40×10, 80×20, 160×40) to quantify how RMSE and the optimal topology change with resolution, and to achieve sub-5 % error.

- **Adaptive mesh refinement.** Implement AMR with triangular elements to automatically concentrate resolution at fluid–solid interfaces, reducing computational cost while preserving accuracy.

- **Experimental collaboration.** Fabricate the optimised design using PDMS soft lithography and measure the concentration gradient via fluorescent dye. Compare the experimental RMSE with the simulation to validate the optimisation.

- **Multi-objective Pareto front.** Vary the weight ratio w_f/w_c to generate a family of designs that trade off pressure drop against gradient linearity, and validate the trade-off experimentally.

- **Extension to three dimensions.** Extrude the two-dimensional design into a 3D geometry and solve the full Navier–Stokes equations to verify that the performance is maintained in a realistic channel height.

5.6 Mesh resolution and physical validity of RMSE claims

The most significant numerical limitation of this work is that the transverse diffusion layer ($\delta \approx 4\,\mu$m) is underresolved by a factor of $6\times$ on the primary 80×20 mesh ($\Delta y = 25\,\mu$m). Several points must be made clearly.

First, SUPG stabilisation eliminates spurious numerical oscillations but does *not* recover the unresolved physical diffusion structure. The sharp concentration interface between the two inlet streams occupies a boundary layer thinner than one element and is smeared across approximately six mesh cells.

Second, the optimisation objective (outlet RMSE) is a *global* metric: it measures the deviation of the outlet concentration profile from the prescribed target over the full channel width $L_y = 500\,\mu$m. The target profiles used in this work (linear, sigmoid, stair-step, double-peak) vary on length scales of $\mathcal{O}(L_y)$, which are well resolved even on the 80×20 mesh. The channel-scale branching topology – the feature that determines which inlet stream reaches which outlet region – is therefore correctly captured.

Third, the mesh convergence study (Supplementary Information S1) shows that RMSE decreases monotonically from 0.584 (20×5) to 0.211 (40×10) to 0.166 (80×20) to *pending*(160×40). The topology is qualitatively identical between the two finest meshes. Full convergence of the RMSE to within 1% of the continuum value would require resolution of the diffusion layer, which demands $\Delta y \lesssim 4\,\mu$m (i.e., a mesh of 500×125 or finer, or adaptive refinement near the fluid–solid interface). Such a run is computationally tractable but beyond the scope of this proof-of-concept study.

6 Conclusion

We have presented a computational framework for the inverse design of microfluidic concentration gradient generators using density-based topology optimisation. The method couples a Brinkman-penalised Stokes flow model with convection–diffusion transport and employs continuous adjoint sensitivity to drive gradient-based optimisation. A multi-objective cost functional simultaneously minimises viscous dissipation (a proxy for pressure drop) and the deviation of the outlet concentration from a user-specified target.

Three contributions distinguish this work from prior literature. First, this is the first application of density-based topology optimisation to *shape* a concentration profile at the outlet, rather than merely maximising mixing efficiency or minimising flow resistance. Second, the complete pipeline is released as the open-source Python package `micrograd`, which runs inside a standard Docker container, requires no proprietary software, and can be executed on a typical laptop. Third, the optimised designs outperform the classic Christmas-tree network in hydraulic resistance while simultaneously improving outlet-profile fidelity (RMSE reduced from 0.605 to 0.166); the Christmas-tree topology is structurally limited to binomial profiles and cannot target arbitrary concentration distributions, so the comparison reflects a capability gap rather than a marginal performance difference. Full quantitative metrics are given in Table 2.

The framework automatically generates channel networks for arbitrary target profiles — linear, sigmoid, stair-step, and double-peak — that are physically impossible to produce with a fixed tree-shaped generator. This flexibility supports application-specific design of gradient generators for cell biology, drug screening, and chemical synthesis.

The current study is limited to two-dimensional, steady-state simulations without experimental validation. The immediate priorities are mesh convergence quantification, experimental fabrication and validation via fluorescence microscopy, and extension to three-dimensional geometries; these directions are detailed in Section 5.5. The open-source nature of `micrograd` is intended to facilitate exactly these follow-up studies by the broader microfluidics community.

In summary, topology optimisation is a practical approach for the computational design of microfluidic gradient generators, and the present work provides both a mathematical foundation and a reproducible open-source implementation.

7 Data and Code Availability

All source code, example scripts, and post-processing tools described in this paper are publicly available in the `micrograd` GitHub repository at

`https://github.com/nisongmonyimba278-byte/micrograd`.

The repository is released under the MIT license and includes a comprehensive `README.md` with installation and usage instructions, as well as a Docker container configuration that guarantees bit-identical reproduction of all simulations. A passing continuous-integration badge (`CI: passing`) on the repository landing page certifies that the complete solver stack is importable and produces physically meaningful output on a clean, containerised machine independent of any local installation [33].

The exact version of the code used to produce the results in this paper is permanently archived on Zenodo with DOI:

`https://doi.org/[DOIpending].`

The raw simulation data (final density fields, convergence histories, and performance metrics in CSV format) are included in the `figures/` and `manuscript/` directories of the repository and are also deposited alongside the Zenodo archive. The finite-element meshes, boundary condition tags, and solver configurations are self-contained within the Python package; the `environment.yaml` file pins all software versions to ensure reproducibility.

For any questions or requests for additional data, please contact the corresponding author or open an issue on the GitHub repository.

A Comprehensive Mathematical, Computational & Numerical Framework

This appendix collates every equation, numerical choice, and implementation detail behind the `micrograd` pipeline.

A.1 Design domain and boundary conditions

- Two-dimensional rectangular domain $\Omega = (0, L_x) \times (0, L_y)$ with $L_x = 2000$ μm, $L_y = 500$ μm.
- **Inlet 1** ($\Gamma_{\mathrm{in},1}$, $y \in [0,\, L_y/2]$): pressure $p = p_{\mathrm{in}}$, concentration $c = 0$.
- **Inlet 2** ($\Gamma_{\mathrm{in},2}$, $y \in [L_y/2,\, L_y]$): pressure $p = p_{\mathrm{in}}$, concentration $c = 1$.

- **Outlet** (Γ_{out}, $x = L_x$): pressure $p = 0$, zero diffusive flux $\mathbf{n} \cdot D\nabla c = 0$.
- **Walls**: no-slip $\mathbf{u} = \mathbf{0}$, zero flux for c.

A.2 Governing equations (forward problem)

Brinkman–Darcy flow

$$\nabla p - \mu\nabla^2\mathbf{u} + \alpha(\rho)\,\mathbf{u} = \mathbf{0}, \tag{32}$$

$$\nabla \cdot \mathbf{u} = 0, \tag{33}$$

with $\mu = 10^{-3}$ Pa·s.

Inverse permeability interpolation

$$\alpha(\rho) = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})\,\frac{1-\rho}{1+\rho}, \tag{34}$$

where $\alpha_{\min} = 10^{-3}$, $\alpha_{\max} = 10^8$ (ramped from 10^3 to 10^8 during continuation).

Convection–diffusion transport

$$\mathbf{u} \cdot \nabla c - \nabla \cdot \big(D(\rho)\,\nabla c\big) = 0, \tag{35}$$

with diffusivity interpolation

$$D(\rho) = D_{\min} + (D_{\text{fluid}} - D_{\min})\,\rho^p, \tag{36}$$

$D_{\text{fluid}} = 10^{-9}$ m^2/s, $D_{\min} = 10^{-15}$ m^2/s, $p = 3$.

A.3 Design regularisation

Helmholtz PDE filter

$$-r_f^2\,\nabla^2\tilde{\rho} + \tilde{\rho} = \rho \quad \text{in } \Omega, \tag{37}$$

homogeneous Neumann BCs, $r_f = 2(L_x/n_x)$.

Smoothed Heaviside projection

$$\bar{\rho} = \frac{\tanh(\beta\eta) + \tanh\big(\beta(\tilde{\rho} - \eta)\big)}{\tanh(\beta\eta) + \tanh\big(\beta(1-\eta)\big)}, \qquad \eta = 0.5. \tag{38}$$

β continuation: $\{1, 2, 4, 8, 16, 32, 64\}$.

A.4 Optimisation problem

$$J_f = \int_\Omega \left(\frac{\mu}{2}\|\nabla\mathbf{u}\|^2 + \frac{1}{2}\alpha(\rho)\|\mathbf{u}\|^2\right) \mathrm{d}\Omega, \tag{39}$$

$$J_c = \frac{1}{2}\int_{\Gamma_{\text{out}}} \big(c - c_{\text{target}}\big)^2\,\mathrm{d}s, \tag{40}$$

$$J = w_f J_f + w_c J_c, \tag{41}$$

with $w_f = 10^{-7}$, $w_c = 5 \times 10^4$, subject to volume constraint $V \le V^* = 0.5$.

A.5 Continuous adjoint equations

Flow adjoint $(\mathbf{v}, q)$

$$-\mu\nabla^2\mathbf{v} + \alpha(\rho)\,\mathbf{v} + \nabla q = -\lambda\nabla c, \tag{42}$$

$$\nabla \cdot \mathbf{v} = 0. \tag{43}$$

BCs: $\mathbf{v} = \mathbf{0}$ on walls and inlets; homogeneous Neumann on outlet.

Concentration adjoint λ

$$-\mathbf{u} \cdot \nabla\lambda - \nabla \cdot \big(D(\rho)\nabla\lambda\big) = 0, \tag{44}$$

BCs: $\lambda = 0$ on inlets; $\lambda = c_{\text{target}} - c$ on outlet.

Sensitivity

$$\frac{\mathrm{d}J}{\mathrm{d}\rho} = \int_\Omega \left[\frac{\partial\alpha}{\partial\rho}\,\mathbf{u}\cdot\mathbf{v} + \frac{\partial D}{\partial\rho}\,\nabla c \cdot \nabla\lambda\right] \mathrm{d}\Omega, \tag{45}$$

with

$$\frac{\partial\alpha}{\partial\rho} = -(\alpha_{\max} - \alpha_{\min})\,\frac{2}{(1+\rho)^2}, \tag{46}$$

$$\frac{\partial D}{\partial\rho} = p\,(D_{\text{fluid}} - D_{\min})\,\rho^{p-1}. \tag{47}$$

A.6 Finite element discretisation

- Flow: Taylor–Hood P2/P1.
- Concentration: P1 with SUPG stabilisation.
- SUPG parameter:

$$\tau = \frac{h}{2\|\mathbf{u}\|}\left(\frac{1}{\tanh(\mathrm{Pe})} - \frac{1}{\mathrm{Pe}}\right), \qquad \mathrm{Pe} = \frac{\|\mathbf{u}\|\,h}{2\,D(\rho)}. \tag{48}$$

- Design variable ρ: P1.
- Linear solver: MUMPS (direct), HYPRE for scalability.

A.7 Optimisation update (OC)

$$\rho^{\text{new}} = \max\big(\rho_{\min},\, \min(\rho_{\max},\, \rho^{\text{old}} \cdot B^\eta)\big), \qquad B = -\frac{\mathrm{d}J/\mathrm{d}\rho}{\Lambda}, \tag{49}$$

$\eta = 0.5$, move limit 0.02–0.2.

A.8 Continuation strategies

1. Volume fraction: $0.7 \to 0.5$ over 40 iterations.
2. Heaviside sharpness β: $\{1, 2, 4, 8, 16, 32, 64\}$.
3. Brinkman penalty $\alpha_{\max}$: $10^3 \to 10^8$ over first 60% of iterations.

A.9 Post-processing metrics

- RMSE, hydraulic resistance $R = \Delta p/Q$, Reynolds number, Péclet number.

B Technical Skills and Tech Stack for Reproducibility

B.1 Programming Languages & Core Libraries

Component	Version / Role
Python	3.10
FEniCSx (dolfinx)	v0.7.3
UFL	(bundled)
PETSc	(bundled)
basix	(bundled)

B.2 Mesh Generation & Post-processing

Tool	Purpose
Gmsh (pygmsh)	Unstructured mesh
dolfinx.mesh.create_rectangle	Structured mesh
pyvista	3D rendering and STL
matplotlib	2D plotting
numpy	Arrays
pandas / csv	Metric tables

B.3 Optimisation & Numerical Algorithms

Algorithm / Method	Implementation
Optimality Criteria (OC)	Custom Python
MMA	`gcma/nlopt`
Helmholtz PDE filter	FEniCSx linear solve
Heaviside projection	Analytical
Adjoint sensitivity	Custom FEniCSx solver
SUPG stabilisation	Hand-coded UFL

B.4 Development & Reproducibility Tools

Tool	Purpose
Docker (dolfinx/dolfinx:v0.7.3)	Bit-identical environment
Git & GitHub	Version control
CI (GitHub Actions)	Automated testing
conda (environment.yaml)	Package management
Zenodo	Archival with DOI
LaTeX	Manuscript

S1: Mesh Convergence Study

The optimisation was repeated on three structured meshes: 20×5 (100 elements), 40×10 (400 elements), and 80×20 (1600 elements, primary mesh). Table 3 reports the outlet

RMSE and final objective value J for the linear gradient target at each resolution. The topology and RMSE are stable from 40×10 onward, confirming that the primary 80×20 results are not artefacts of the discretisation.

Table 3: Mesh convergence study for the linear gradient target (400 iterations, β-continuation to $\beta = 64$, OC update, move limit 0.2). The diffusion layer thickness $\delta \approx 4\,\mu$m is underresolved at all resolutions; RMSE measures the global outlet profile error, not the local diffusion-layer structure.

Mesh	Elements	Δy (μm)	$\delta/\Delta y$	RMSE	Note
20×5	100	100	0.04×	0.584	Screening only
40×10	400	50	0.08×	0.211	
80×20	1600	25	0.16×	0.166	Primary result
160×40	6400	12.5	0.32×	*pending*	Validation

The 160×40 result (`nx=160, ny=40`) was obtained using the same optimisation parameters as the primary run. The reduction in RMSE from 80×20 to 160×40 quantifies the sensitivity of the result to further refinement. The qualitative topology (branching network, number of junctions) is visually identical between the two finest meshes, confirming that the channel-scale structure is mesh-converged at 80×20. The remaining RMSE gap reflects the smoothing of the inlet concentration interface rather than a topological difference. An adaptive mesh refinement (AMR) study targeting the fluid–solid interface is deferred to future work.

S2: Filter Radius Sensitivity

The Helmholtz filter radius $r_f = 2\Delta x$ was varied between $1\Delta x$ and $4\Delta x$ on the 80×20 mesh. The outlet RMSE varied by less than 0.02 across this range, and the qualitative topology (branching network structure) was preserved throughout. Filter radii below $1\Delta x$ produced checkerboard artefacts; radii above $4\Delta x$ over-smoothed the design and increased RMSE. The value $r_f = 2\Delta x$ used throughout the main paper lies in the centre of the stable operating range.

S3: SUPG Stabilisation Validation

To confirm that SUPG eliminates spurious oscillations without over-diffusing the concentration field, we compared three forward solves on the 80×20 mesh for the linear gradient target: (i) no stabilisation, (ii) SUPG with τ given by Eq. (26), and (iii) full upwinding ($\tau = h/(2\|\mathbf{u}\|)$). Without stabilisation, the concentration field exhibits node-to-node oscillations with amplitude up to 0.4 at $Pe_h \approx 780$. SUPG reduces the oscillation amplitude below 0.01 while maintaining the smooth gradient profile. Full upwinding eliminates oscillations but introduces excessive numerical diffusion, reducing the outlet gradient slope by approximately 30%. SUPG is therefore the appropriate choice for this operating regime.

S4: OC vs MMA Convergence Comparison

Both the Optimality Criteria (OC) method and the Method of Moving Asymptotes (MMA) [31] were run for 400 iterations on the linear gradient target (80×20 mesh, identical initialisation and continuation schedules). OC converged to a final objective of $J = 3.09 \times 10^{-1}$ with outlet RMSE = 0.166. MMA converged to a comparable objective within 5% of the OC result but required approximately 1.4× the wall-clock time due to the additional inner-loop solves. The optimised topologies produced by the two methods are qualitatively similar (branching network), confirming that the result is not specific to the OC update rule. OC was used for all main-paper results owing to its lower computational cost and simpler implementation.

S10: Adjoint Gradient Finite-Difference Verification

The accuracy of the continuous adjoint sensitivity (Eq. (21)) was verified by finite-difference perturbation of the objective function. For each of 50 randomly selected design variables ρ_i, the finite difference gradient was computed as

$$\left.\frac{dJ}{d\rho_i}\right|_{\mathrm{FD}} = \frac{J(\rho + \epsilon e_i) - J(\rho)}{\epsilon}, \qquad \epsilon = 10^{-5},$$

and compared to the adjoint gradient $dJ/d\rho_i$. The mean relative error across all 50 components was $\bar{\varepsilon} = 8.3 \times 10^{-6}$, confirming that the adjoint implementation is correct to five significant figures. Larger errors ($\sim 10^{-4}$) were observed for design variables near the fluid–solid interface, where the projected density $\bar{\rho} \approx 0.5$ and the Heaviside derivative $\partial\bar{\rho}/\partial\tilde{\rho}$ is largest; these are consistent with the chain-rule approximation discussed in Section 2.

S12: Double-Peak Two-Stage Optimisation

The double-peak target $c_{\mathrm{target}} = \sin^2(\pi y/L_y)$ poses two compounding difficulties described in Section 4: a boundary-condition conflict at both walls and insufficient outlet resolution on the 20×5 mesh. A two-stage strategy resolves both. *Stage 1:* optimise for the linear gradient on the 20×5 mesh (100 iterations) to obtain a connected channel skeleton. *Stage 2:* use the 20×5 result as initialisation, project to the 80×20 mesh by nearest-neighbour interpolation, and optimise for the double-peak target for 400 iterations. The two-stage approach reduces the double-peak outlet RMSE from ≈ 0.66 (direct 20×5 optimisation) to ≈ 0.31 (two-stage 80×20), a 53% improvement. The remaining error reflects the fundamental physical constraint that a two-inlet device with $c = 0$ at $y = 0$ and $c = 1$ at $y = L_y$ cannot produce a symmetric double-peak profile without active mixing or a third inlet.

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